

Herald



Herald — Continuation

Field	Value
Revision	12
Created	2026-05-20
Last modified	2026-05-27
Status	active
Status summary	**2026-05-27 GAP-3 + HRD-132 close-out checkpoint (HEAD 4f45f26; constitution at 3c9c4e9). GAP-3 §11.4.85 stress+chaos suite (HRD-122..HRD-130) AND the HRD-132 Runner idempotency-window fix are COMPLETE and now atomically migrated Issues→Fixed (Issues.md r16 / Fixed.md r15, §11.4.19): HRD-122..HRD-130 Task→Completed, HRD-132 Bug→Fixed. HRD-132 fix 354b883 lands claim-before-dispatch at Runner Stage 2 — INSERT events_processed ... ON CONFLICT DO NOTHING is the AUTHORITATIVE exactly-once dispatch gate (the Stage-2→Stage-7 duplicate-dispatch window HRD-125 surfaced is closed); latent runner .go: 132 CachedRcpt .WasReplay race fixed; HRD-125 assertion upgraded to sends==1 (docs/qa/HRD-125-stress-chaos-2026-05-27T131251-gap3/ shows dispatch_exactly_once=1, shared_key_sends=1 want=1); wave3 mutation-gate M4 retired (4f45f26) — Stage-7 Insert now redundant. Full scripts/e2e_bluff_hunt.sh 78 PASS / 0 FAIL; HRD-130 gate 6 PASS / 0 FAIL (5 load-bearing mutations). HRD-131 (SQLite SSoT migration Phase 1) remains OPEN; Phases 2-6 follow-on. Prior GAP-3 checkpoint (HEAD d2a60d3, all 4 mirrors synced): GAP-3 §11.4.85 stress+chaos suite COMPLETE (commits 02d272d→d2a60d3, HRD-122..130): HRD-122 scaffold commons/ stresschaos/, HRD-123 Gin /v1/events, HRD-124 /v1/ compliance+/v1/safety_state, HRD-125 Runner, HRD-126 pherald listen, HRD-127 claude_code (testdata fake-claude shims), HRD-128 container/resource-exhaustion (§12.6-safe, live OOM SKIP-with-reason), HRD-129 e2e E81-E88, HRD-130 paired §1.1 mutation gate tests/ test_stress_chaos_mutation_meta.sh. Full scripts/ e2e_bluff_hunt.sh now reports 78 PASS / 0 FAIL (was 60; +18 = E81-E88 + anchor sub-checks); HRD-130 gate 6 PASS / 0 FAIL (5 mutations load-bearing). Evidence committed under docs/qa/HRD-12{3,4,5,7,8} - * + docs/qa/ 2026-05-27T135301-gap3-listen-HRD-126/. NOTE — HRD-122..130 are now MIGRATED to Fixed.md (Completed) (suite landed +

Field	Value
	e2e green + atomic Issues→Fixed migration done in Issues.md r16 / Fixed.md r15). Two anti-bluff issues caught + fixed mid-session by the conductor: HRD-123 flaky oversized-body assertion (§11.4.50 — transport-rejection of an 8 MiB body is valid, not a failure) and E82 stale-anchor (re-captured committed evidence with HERALD_STRESS_QA_DIR; assertion <code>all_4xx_no_panic→all_malformed_rejected_no_5xx</code>). HRD-132 (FIXED 354b883, now in Fixed.md): Runner idempotency window — concurrent same-key replay duplicate-dispatched (2-4 of a 1000× flood) because <code>events_processed</code> was archived at Stage 7 while <code>idempotency.go:64-72</code> treated SETNX-miss+no-PG-row as fresh (documented §32.1 "Redis-lies-PG-truths"). FIX landed claim-before-dispatch (INSERT <code>events_processed ... ON CONFLICT DO NOTHING</code> at Stage 2 = authoritative exactly-once dispatch gate) + closed the latent runner <code>.go:132</code> <code>CachedRcpt.WasReplay</code> race + upgraded the HRD-125 assertion to <code>sends==1</code> (evidence <code>dispatch_exactly_once=1</code>); wave3 M4 retired (4f45f26). Prior r10 (still valid history) — a constitution-compliance + SSoT-hygiene sweep: (1) pulled HelixConstitution 3a085b9→3c9c4e9 (gained §11.4.89 background-test / §11.4.90 Obsolete-status / §11.4.91 summary-clarity / §11.4.92 multi-pass-eval / §11.4.93 SQLite-SSoT / §11.4.94 zero-idle-parallel); (2) propagated §11.4.89-94 into CLAUDE.md r12 / AGENTS.md r8 / HERALD_CONSTITUTION.md r6 §108.d-i (commit 83b5e2f) — inheritance gate 15 PASS / 0 FAIL; (3) built <code>scripts/mutation_residue_audit.sh</code> §107.y pre-push residue scanner (e23933c) + back-filled §107.y quiescence + <code>.git/MUTATION_IN_PROGRESS</code> lockfile into wave2/3/4 gates (c19a092); (4) re-verified ALL mutation gates green one-at-a-time — wave2 5/0, wave3 6/0, wave4 5/0, wave4b 5/0, wave6.5 6/0 (wave6 already 4/4), each with Quiescence assertion PASS; (5) §11.4.91 summary-clarity — 7 one-liners fixed (d24e829); (6) operator correction — <code>docs/.workable_items.db</code> is VERSION-CONTROLLED, NOT gitignored (Herald divergence from parent §11.4.93, 30f9c6d); (7) GAP-3 §11.4.85 stress+chaos PLAN committed (a422c53, HRD-122..130); (8) read-only audits committed (9c09496 constitution-compliance, ee63ce6 HRD-obsolescence + qa-coverage); (9) HRD-131 filed (SQLite SSoT Phase 1) in Issues.md. RESOLVED debts: Wave 6.5 mutation-gate M3 stale anchor (<code>opts.ReplyTo→textOpts.ReplyTo</code>) FIXED; Postgres-SASL e2e carry-over RESOLVED (e2e now 60 PASS / 0 FAIL / 24 SKIP); HRD-101 now filed as a real Fixed.md row. Prior session: v0.4.0 (Wave 6 closed-loop) + v0.5.0 (Wave 6.5 lifecycle) tagged; Wave 7 T1-T5 closed (HRD-110..114). PENDING-OPERATOR: MTPROTO real-channel automation (Telegram never relays bot-to-bot in groups — <code>docs/research/telegram-bot-to-bot-constraint.md</code>, 5267f14; needs <code>my.telegram.org</code> app_id+app_hash + / <code>revoke @pherald_qa_bot</code>).
Issues	HRD-008, HRD-015, HRD-018 (in progress), HRD-019..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090, HRD-115..HRD-121 (Wave 7 T6-T12 pending), HRD-131 (SQLite SSoT Phase 1, open)
Issues summary	see Issues.md. Wave 7 closed HRD-110/111/112/113/114; HRD-115..121 are the remaining Wave 7 tasks T6-T12. HRD-122..HRD-130 (§11.4.85 stress+chaos suite, GAP-3) + HRD-132 (Runner idempotency-window bug) are now MIGRATED to Fixed.md (Completed / Fixed) — Issues.md r16 / Fixed.md r15 atomic §11.4.19 migration. HRD-131 is §11.4.93 SQLite workable-items SSoT migration Phase 1 (open) — Phases 2-6 follow-on.
Fixed	HRD-001..HRD-017 (less open), HRD-028, HRD-080, HRD-092..HRD-101, HRD-110..HRD-114, HRD-122..HRD-130, HRD-132

Field	Value
Fixed summary	see Fixed.md Resume — remaining "do everything" workstreams in priority order (GAP-3 + HRD-132 DONE): (1) §11.4.93 SQLite SSoT migration Phase 2+ (HRD-131, open): Phase 2 adopt the constitution's migration binary (constitution/scripts/workable-items/, §11.4.74 reference — never reimplement), then Phases 3-6 (md→db sync, db→md byte-identical regen, generator shims, text-direct-edit prohibition) opened as follow-on HRDs so §106/§107.x invariants never break mid-migration. (2) v1.0.0 readiness — 39 open §42/§43 HRDs (HRD-018..027, 029..056, 081, 085..090, 008, 015) + Wave 7 T6-T12 (Slack adapter via vendored slack-go submodule — verify I6 gate after .gitmodules entry, confirm slack-go v0.16.0 field names inner.Files/socketmode.EventTypeEventsAPI/UploadFileV2Parameters; then qaherald Slack MessengerClient, spec §11.0/§32.2/§43, e2e, mutation gate, Issues→Fixed/v0.6.0/4-mirror push; T5 pre-staged perChannelConfig("slack") + blank-import in listen.go — HERALD_CHANNELS=tgram, slack currently errors ErrUnknownChannel until T6 registers slack, by design; plan docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md line 980). (3) GAP-4 docs/ Continuation qa back-fill + docs-audit #147 — features lacking docs/qa/<run-id>/evidence per the ee63ce6 qa-coverage audit + comprehensive documentation audit. STILL-OPEN GATED ITEMS: MTProto real-channel automation — OPERATOR-DECISION: provide my.telegram.org app_id+app_hash (folds into a qaherald MessengerClient MTProto impl reusing the entire scenario engine; Telegram never relays bot-to-bot in groups — see docs/research/telegram-bot-to-bot-constraint.md); operator should also /revoke @pherald_qa_bot (leaked plaintext token). §11.4.90 obsolescence — zero current Obsolete rows; convention in place, periodic re-audit pending. (RESOLVED this session: GAP-3 §11.4.85 stress+chaos suite IMPLEMENTED — e2e 78 PASS / 0 FAIL — AND atomically migrated Issues→Fixed (HRD-122..130 Completed); HRD-132 Runner idempotency-window bug FIXED 354b883 (claim-before-dispatch, exactly-once sends==1) + migrated to Fixed.md; HRD-123 flaky oversized-body assertion + E82 stale-anchor anti-bluff issues fixed. Prior r10 RESOLVED: Wave 6.5 mutation-gate M3 stale anchor; Postgres-SASL e2e FAILs; HRD-101 filed as a real Fixed.md row.) §11.4.87 endless-loop: continue until Issues.md zero-active + CONTINUATION §3 empty + no subagent in flight. NO BLUFF. Push 4 mirrors every 2-3 commits.

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§0. How to use this document

Paste the following block into any CLI agent (Claude Code / OpenCode / Cursor / Aider / Gemini CLI) to resume Herald work exactly where it was left:

You are working on the Herald project at ~/Projects/Herald (also reachable as the Herald/ submodule of a consuming project). The Helix Universal Constitution lives at <ancestor>/constitution/ (parent-walk discovery). Read in this order: CLAUDE.md, AGENTS.md, README.md, docs/guides/HERALD_CONSTITUTION.md, docs/guides/CONSTITUTION_INHERITANCE.md, docs/specs/mvp/specification.V3.md. Then read docs/CONTINUATION.md (this file) for live state, docs/Issues.md for open work, docs/Status.md for current phase, docs/Fixed.md for closed history. Go workspace builds via go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/.... Inheritance gate tests/test_constitution_inheritance.sh MUST exit 0 before any commit. Multi-mirror fan-out push to four hosts (GitHub + GitLab + GitFlic + GitVerse) is mandatory per Constitution §103.

§1. Snapshot

- **Active spec:** docs/specs/mvp/specification.V3.md Revision 4 (~4300 lines).
- **Archived specs:** V1 + V2 in docs/specs/mvp/archive/ (frozen).
- **Go modules:** commons, commons_prefix, commons_messaging, commons_storage, pherald (all compile + unit tests pass).
- **Container scaffold:** quickstart/{docker-compose.quickstart.yml, Dockerfile.pherald, otel-config.yaml, .env.example} for §26.5 quickstart.
- **CLI:** pherald version --json returns canonical build info; serve/send/doctor/migrate/subscriber/deadletter stubbed with HRD-NNN pointers.
- **Inheritance gate:** 15 PASS / 0 FAIL. Meta-test ✓.
- **e2e_bluff_hunt:** 78 PASS / 0 FAIL (was 60; +18 = GAP-3 E81-E88 §11.4.85 stress+chaos + anchor sub-checks). Plus SKIP-with-reason for hardware/credential-absent invariants (live OOM, MTProto).
- **Stress+chaos mutation gate:** tests/test_stress_chaos_mutation_meta.sh (HRD-130) 6 PASS / 0 FAIL (5 load-bearing mutations).
- **Phase:** implementation r1.

§2. Last commit landed

This commit (V3 r4 + Go scaffold + tracking-doc refresh) closes HRD-009/HRD-009b/HRD-013/HRD-014 (with a Universal §11.4.19 atomic Issues.md → Fixed.md migration) and lands spec V3 §37–§41 (tracker events, workable-item announcement contract, message presentation + Herald Canonical Template, docs/tests completeness, REST API surface). Builds + tests:

```
$ go test ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_storage/...
ok      github.com/vasic-digital/herald/commons 1.135s
ok      github.com/vasic-digital/herald/commons_prefix 0.639s
ok      github.com/vasic-digital/herald/commons_messaging/channels/null 0.000s
ok      github.com/vasic-digital/herald/commons_messaging/dispatch/claude 0.000s
ok      github.com/vasic-digital/herald/commons_storage 1.630s
```

§3. Active work

ID	Status	What
HRD-008	in_progress	Operator-side Quickstart compose validation (Postgres + Redis + OTel + pherald container) — scaffold shipped this commit; live end-to-end run pending operator.
HRD-010	open	commons_storage live (pgx pool + golang-migrate driver + River queue + Redis ACL).
HRD-011	open	Telegram adapter live (telebot SDK + getUpdates long-poll + webhook secret_token).
HRD-012	open	Claude Code dispatcher live (c l a u d e --resume + parse <<<HERALD-REPLY>>>).
HRD-015	open	Inheritance gate I8 invariants for Go scaffold (go.work + commons/types.go + null adapter test passes).
HRD-016	open	REST API surface via Gin Gonic per spec §41.
HRD-017	open	Propagate Universal §11.4.6X spec-versioning + submodule-catalogue-first mandates into the parent constitution.
HRD-122..130	Completed	GAP-3 §11.4.85 stress+chaos suite — IMPLEMENTED + e2e-green (d2a60d3); MIGRATED Issues→Fixed (Fixed.md r15). DONE.
HRD-131	open	§11.4.93 SQLite workable-items SSoT migration Phase 1 (scope-lock filed); Phase 2+ pending. ← the only remaining open item filed this GAP cycle.
HRD-132	Fixed	Runner idempotency-window bug — FIXED 354b883 (claim-before-dispatch at Stage 2, exactly-once sends==1, CachedRcpt.WasReplay race closed); MIGRATED Issues→Fixed (Fixed.md r15). DONE.

§4. Next concrete steps

1. **HRD-018 Catalogue-Check survey + commons_constitution scaffold** — first action of the §42 implementation rollout. **MUST start** with a vasic-digital + HelixDevelopment catalogue survey per the brand-new Universal §11.4.74. Record Catalogue-Check: reuse|extend|no-match on HRD-018 before any Go code lands. Then scaffold the Evaluator interface, 12 event-class emit helpers, constitution_state + constitution_bindings migrations, bundle-hash captureer, mode-ladder runtime config.
2. **HRD-008 quickstart validation** — On a fresh laptop with Podman or Docker:

```
git clone <Herald repo>
git submodule update --init
cd quickstart
podman build -t herald/pherald:dev -f Dockerfile.pherald ../../
cp .env.example ../../.env && $EDITOR ../../.env
podman-compose -f docker-compose.quickstart.yml up -d
curl --retry 30 --retry-delay 2 http://localhost:24090/readyz
```

The current pherald serve returns "not implemented" — HRD-010/HRD-011/HRD-012 must land first to make the live curl POST /v1/events succeed end-to-

end. Validation reveals which of the spec's assumptions (port ranges, Compose syntax, OTEL collector config, Postgres healthcheck) hold against real infrastructure.

3. **HRD-010** — wire `commons_storage/storage.go`'s `MigrationDriver` to `golang-migrate`; bring `pgx` + `River` + `Redis` client up; add integration tests under `// go:build integration`.
4. **HRD-011** — replace `commons_messaging/channels/tgram/tgram.go` stub with a live implementation against `gopkg.in/telebot.v3` or `github.com/mymmrac/telego`; recorded HTTP fixtures under `testdata/`.
5. **HRD-012** — replace `commons_messaging/dispatch/claude_code/claude_code.go`'s `Dispatch` stub with a real `claude --resume` invocation; capture session UUID; parse `<<<HERALD-REPLY>>>`.
6. **HRD-016** — scaffold `pherald/internal/http/` with Gin routes per V3 §41; wire `pherald serve` to mount the Gin router on `http_port`.
7. **HRD-017** — propagate Universal §11.4.6X new mandates (spec-versioning + submodule-catalogue-first) into the constitution submodule.

§4a. Wave 6 live-test handoff (T10b — gating tag v0.4.0)

Operator-supplied credentials required — Wave 6's e2e invariants E63-E70 land as honest SKIPs until the live closed-loop runs against real Telegram + real Claude Code with real chat messages exchanged.

Step-by-step.

1. Export credentials in the shell that will drive the test:

```
export HERALD_TGRAM_BOT_TOKEN="<your-bot-token-from-BotFather>"
export HERALD_TGRAM_CHAT_ID="<numeric-chat-id-where-bot-is-admin>"
export HERALD_CLAUDE_BIN="$(command -v claude)"      # path to
the claude CLI binary
export HERALD_PROJECT_NAME="Herald"                  # or any
operator-chosen session name
```

2. Pick a `<run-id>` (timestamp + 4-char nonce):

```
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-$(uuidgen | cut -c1-4)"
QA_DIR="docs/qa/HRD-100-{$RUN_ID}"
mkdir -p "{$QA_DIR}/attachments"
```

3. Type ONE message in the configured Telegram chat (script reads it via `getUpdates` within 60s).
4. Run the live closed-loop test:

```
bash tests/test_wave6_live_loop.sh
```

The script builds pherald, observes the original message_id, starts pherald listen --bot-token "\$HERALD_TGRAM_BOT_TOKEN" --chat-id "\$HERALD_TGRAM_CHAT_ID" --qa-out-dir "\$QA_DIR" in the background, waits up to 45s for a reply with reply_to_message_id == original, and exits 0 on PASS.

5. Capture supplementary logs:

```
cp /tmp/pherald-w6.log "${QA_DIR}/pherald-listen.log"
# claude stdout/stderr — paths depend on the journaling
# setup. Copy whatever
# the test produced into ${QA_DIR}/claude-stdout.log +
# claude-stderr.log.
```

6. Author a brief \${QA_DIR}/README.md (5 lines minimum) narrating: who ran it, when, what message was sent, what reply came back, any anomalies. This is NOT auto-generated — operator-written narrative is the §107.x evidence anchor.

7. git add \${QA_DIR}/ && git commit -m "Wave 6 step 10b: docs/qa/HRD-100-\${RUN_ID}/ live closed-loop evidence" then proceed to T13b (v0.4.0 tag + 4-mirror push).

If the script SKIPS (any env var unset): the test prints SKIP: <reason> and exits 0. That is honest §11.4.3 hardware-absent SKIP-with-reason, not a PASS. Tag v0.4.0 MUST NOT be created until at least one PASS run is committed to docs/qa/HRD-100-<run-id>/.

§4b. qaherald-auto resume runbook (next session — gating tag v0.5.1)

Status as of 2026-05-23 session end:

- HEAD 8ad4cb9 on main; all 4 mirrors in sync (github.com:vasic-digital/Herald.git, gitlab.com:vasic-digital/herald.git, gitflic.ru:vasic-digital/herald.git, gitverse.ru:vasic-digital/Herald.git).
- Tag v0.4.0 shipped (Wave 6 closed-loop). Tag v0.5.0 shipped (Wave 6.5 lifecycle implementations — premature: only S1+S2 live evidence + carries the pre-53a7ad3 envelope-action-guidance bug). Tag v0.5.1 is the target after qaherald-auto T6-T8 land + full 15-scenario evidence committed.
- Plan: docs/superpowers/plans/2026-05-23-qaherald-auto.md (commit 52ce679).
- Done: T1 skeleton + T2 messenger + T3 scenarios + T4 orchestrator + T5 report (combined commit 8ad4cb9 for T3-T5).
- TODO: **T6 pre-flight** + **T7 hermetic tests** + **T8 shell-adaptor** then live run + tag.

Resume prerequisites

The following are **already provisioned** in ~/.zshrc (export-on-shell-start):

```
HERALD_TGRAM_BOT_TOKEN
# @atmosphere_worker_bot — pherald-bot (Telegram bot
# 8823384001)
HERALD_TGRAM_CHAT_ID      # -4946584787 (ATMOSphere Development
# group)
```

```
HERALD_OPERATOR_IDS      # 2057253161 (the operator's user-id)
HERALD_QA_BOT_TOKEN      # @pherald_qa_bot — 2nd bot
                          (8971749017, privacy DISABLED, member of the group)
```

The qa-bot token is plaintext in `~/ .zshrc` — operator should `/revoke` via `@BotFather` after Wave 6.5 close-out and regenerate (the token leaked into one prior session's bash output earlier today, was never committed to git but lives in `/private/tmp/claude-501/...` operator-local files).

The containers `herald-postgres` (port 24100) and `herald-redis` (port 24200) are LIVE — both have been Up since the start of this session; data persists across podman restarts. Schema is at migration v12 (all 12 applied).

Resume sequence (suggested next-session order)

1. Pull state + verify:

```
cd ~/Projects/Herald && git pull origin main && git log --
    oneline -10
git status --short # expect clean
git stash list     # 2 stashes preserved: T8 Wave-5 salvage
                  + premature S3-S5 transcript WIP
```

2. **Spawn T6 (pre-flight)** via a fresh subagent — read `docs/superpowers/plans/2026-05-23-qaherald-auto.md` Task 6 in full. Implementation lands at `qaherald/internal/lifecycle/preflight.go + preflight_test.go`. 10 gates: pherald-bot reachable via `getMe`, qa-bot reachable, qa-bot privacy disabled, qa-bot is group member, `OPERATOR_IDS` contains qa-bot, etc. Each gate has distinct exit code for diagnostics.

3. **Spawn T7 (full hermetic tests)** — extends T3-T5's 12 tests with `httptest`-based scenario simulation. Coverage targets: every scenario PASS path + every FAIL diagnostic + S9 SKIP path + S14 outbound-attachment sha256 round-trip + S11/S12/S13 inbound-attachment download + `Issues.md/Fixed.md` fs-mutation assertion.

4. **Spawn T8 (shell-script adapter)** — rewrites `tests/test_wave6.5_lifecycle.sh` to delegate to `qaherald lifecycle` for automated runs. Keeps the original `--manual` flag for the operator-typing interactive UX as a fallback.

5. Run the live e2e (the §107 watershed):

```
cd ~/Projects/Herald
# Add qa-bot user-id to OPERATOR_IDS so S5/S6/S8/S10 succeed
export HERALD_OPERATOR_IDS="$
    {HERALD_OPERATOR_IDS},8971749017"
# Re-export from .zshrc (or just open a new shell)
source ~/.zshrc

# Build pherald + qaherald
go build -o /tmp/pherald ./pherald/cmd/pherald
go build -o /tmp/qaherald ./qaherald/cmd/qaherald
```



```
# Start pherald listen in background with QA journaling
RUN_ID="$(date -u +%Y-%m-%dT%H-%M-%S)-w6.5live"
PHERALD_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`-pherald"
QAUTO_QA_DIR="docs/qa/HRD-101-lifecycle-`${RUN_ID}`"
mkdir -p "${PHERALD_QA_DIR}/attachments" "${QAUTO_QA_DIR}/
    attachments"
/tmp/pherald listen --qa-out-dir "${PHERALD_QA_DIR}" --docs-
    dir docs &
PHERALD_PID=$!
trap 'kill -TERM $PHERALD_PID 2>/dev/null' EXIT
sleep 5

# Run qaherald lifecycle (T6-T8 must be done first)
/tmp/qaherald lifecycle \
    --pherald-bot-username=atmosphere_worker_bot \
    --pherald-qa-out-dir="${PHERALD_QA_DIR}" \
    --out="${QAUTO_QA_DIR}" \
    --run-id="${RUN_ID}"
# Exit 0 on all-PASS or all-PASS-with-S9-SKIP. Non-zero on
    any FAIL.
```

6. Capture evidence:

- `\${PHERALD\_QA\_DIR}/transcript.jsonl` — pherald's view (classifications, dispatch, replies)
- `\${QAUTO\_QA\_DIR}/transcript.jsonl` — qaherald-auto's view (sends, assertions)
- `\${QAUTO\_QA\_DIR}/report.md` — Markdown summary
- `\${QAUTO\_QA\_DIR}/attachments/<sha256>.<ext>` — content-addressed inbound + outbound attachments
- docs/Issues.md + docs/Fixed.md — mutated by S5/S6/S8/S10 (and reverted by S10 + S15 cleanups; both files end in valid state)

7. Author operator README + commit:

```
cat > "${QAUTO_QA_DIR}/README.md" <<EOF
# HRD-101 Wave 6.5 lifecycle evidence — <run-id>

<5+ lines narrating: who ran it, when, which scenarios
    PASSEd/FAILEd/SKIPped,
bot reply quality observations, any anomalies. Operator-
    written, NOT auto-generated.>
EOF
git add docs/qa/HRD-101-lifecycle-`${RUN_ID}`*
git commit -m "Wave 6.5 step 13: HRD-101 live 15-scenario
    lifecycle evidence (qaherald-auto)"
```

8. Tag v0.5.1 + 4-mirror push:

```
git tag -a v0.5.1 -m "Wave 6.5 close: full lifecycle live
evidence + envelope action-guidance fix"
git push origin main
git push origin v0.5.1
# Verify all 4 mirrors converged on the same SHA + tag
for mirror in github.com:vasic-digital/Herald.git
gitlab.com:vasic-digital/herald.git
gitflic.ru:vasic-digital/herald.git
gitverse.ru:vasic-digital/Herald.git; do
    git ls-remote "git@${mirror}" v0.5.1
done
```

Open / known constraints for tomorrow

- **S9 (non-operator Done: rejection):** requires HERALD_QA_BOT_TOKEN_NON_OPERATOR env (a THIRD bot account whose user-id is NOT in OPERATOR_IDS). Without it, S9 emits a SKIP-with-reason. Operator may register a third bot OR accept the SKIP (S9's logic is already unit-tested hermetically in T5's mutation gate).
- **First-scenario CC bootstrap:** ~30s on first inbound message because the Claude Code session must spawn. Subsequent scenarios use the cached session (~5-15s typical).
- **Issues.md mutation race:** if you run qaherald lifecycle while ANOTHER pherald listen is also processing in the same checkout, Issues.md may receive double-writes. Single-pherald-listen invocation is the safe default.
- **Postgres-SASL carry-over (RESOLVED 2026-05-27):** the e2e_bluff_hunt FAILs on Postgres-SASL invariants (E7-E12 + E14-E16 + E37-E42) are FIXED — 9b166c4 (auth env + PG password self-heal + Redis/tenant wiring + honest SKIP-with-reason) + bdb9f1 (Runner nil-Redis graceful PG-only fallback). At resolution e2e_bluff_hunt reported 60 PASS / 0 FAIL / 24 SKIP; the later GAP-3 session (d2a60d3) raised it to **78 PASS / 0 FAIL** with E81-E88 wired in.

Safe-shutdown checklist (HISTORICAL — end of 2026-05-23 session; superseded — current HEAD is d2a60d3)

Check	State
All committed work pushed to 4 mirrors	YES — HEAD 8ad4cb9 confirmed on github+gitlab+gitflic+gitverse
Local working tree clean	YES — git status --short empty
Stashes preserved	2 stashes: T8 Wave-5 salvage (older) + premature mid-test S3-S5 transcript WIP (newer, will be regenerated cleanly by qaherald-auto in next session)
Background pherald/qaherald processes	NONE running (ps aux grep pherald empty)
Container state	herald-postgres + herald-redis Up 17 hours — persist data across podman stop && podman start; can be left running OR stopped (podman stop herald-postgres herald-redis); next session restart is podman start herald-postgres herald-redis

Check	State
/tmp/*.log files	one operator-token leak shred-deleted earlier today; remaining transient logs contain no credentials (verified)
.zshrc	contains 4 HERALD_* exports including HERALD_QA_BOT_TOKEN plaintext — operator should /revoke + regenerate via @BotFather post-Wave-6.5 closure
Memory entries	All session findings persisted under /Users/milosvasic/.claude/projects/-Users-milosvasic-Projects-Herald/memory/ — survive across sessions

Machine is safe to shut down. Resume tomorrow by `cd ~/Projects/Herald && git pull origin main` and start with `qaherald-auto T6`.

§5. Long-form pointers

- docs/specs/mvp/specification.V3.md — full active spec (Revision 4).
- docs/specs/mvp/specification.V3.md#30-v2-self-review-log — every review pass.
- docs/guides/HERALD_CONSTITUTION.md — §101..§106 extending Universal.
- docs/guides/CONSTITUTION_INHERITANCE.md — parent-discovery + gate.
- tests/test_constitution_inheritance.sh — the gate.
- quickstart/ — HRD-008 scaffold.
- commons/types.go — the §11.0 type contract reference.